Find what moves in the SPHEREx sky.

Planet X is a planet several times Earth’s mass, predicted far beyond Neptune and never seen. Between two pictures of the same sky stars stay put; anything bound to the Sun does not. Muninn turns SPHEREx visits into that comparison, in a browser.

From the NASA Space Apps 2026 challenge “Planet X and SPHEREx”.

The whole sky
every six months

102 channels
0.75–5 µm, the near infrared

Since May 2025
surveying, release after release

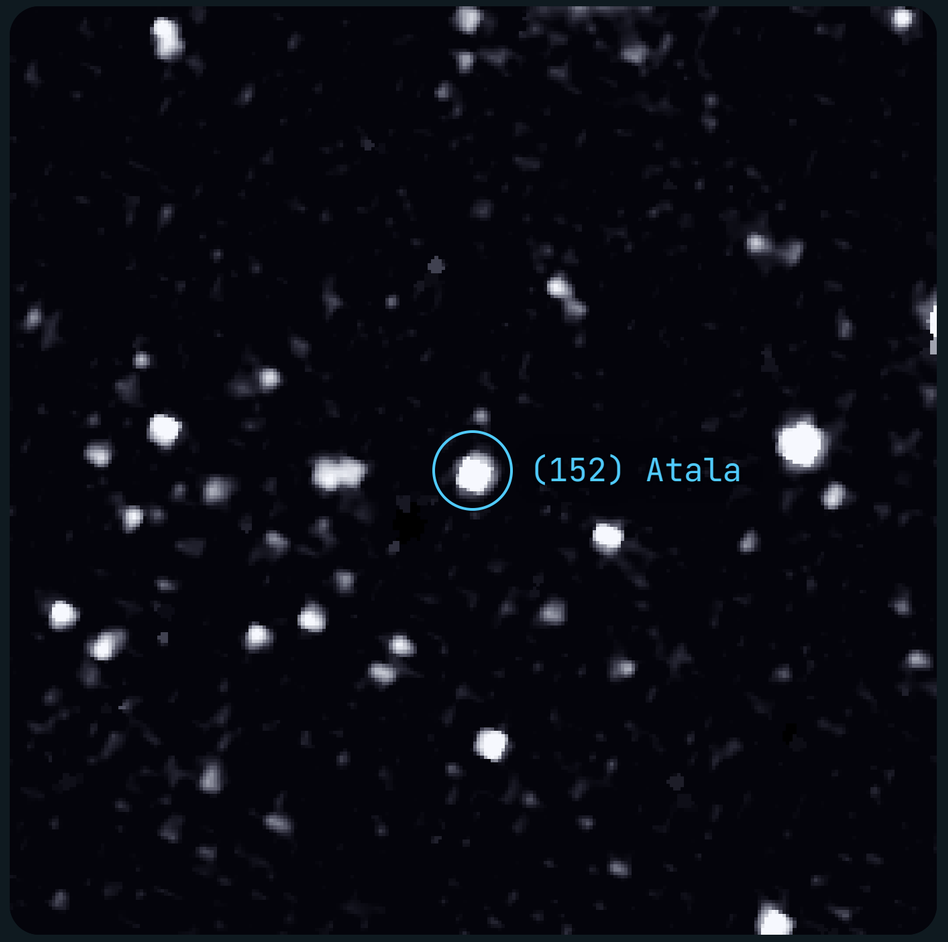
Who it is for

- The public: one link, nothing to install
- Volunteers in the Backyard Worlds style, blinking for movers
- Teachers: real NASA frames in class

See what changed between two SPHEREx visits.

Muninn puts the visits of one field on one grid and blinks them: stars hold still, a mover jumps.

These frames are being read from the NASA archive right now, in this browser.



25 Dec 2025

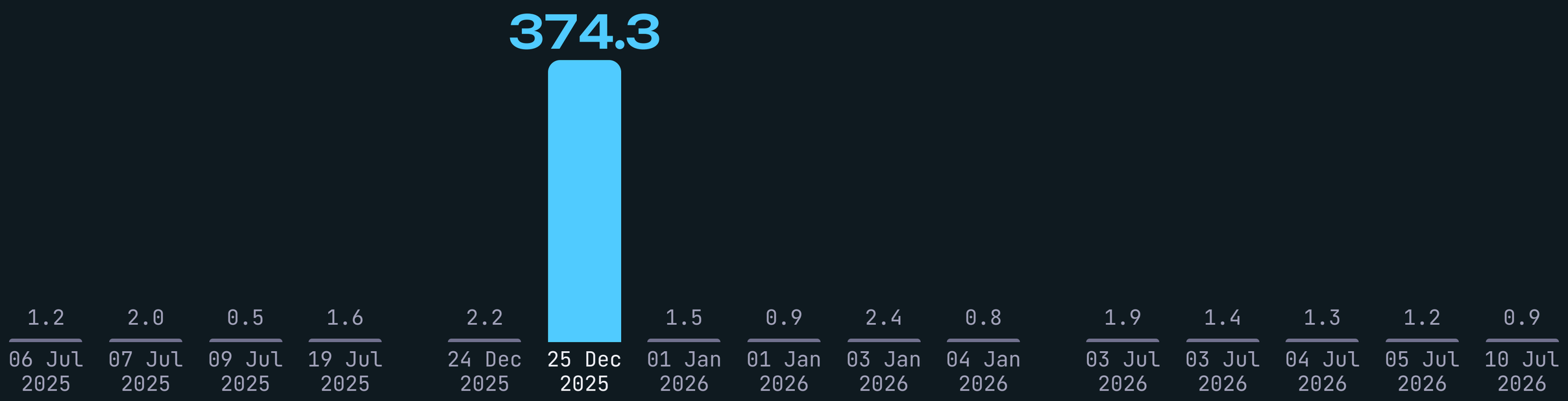
D2 · 1.10–1.65 μm · 9′ × 9′

24 Dec 2025	1.6 MB
25 Dec 2025	1.6 MB
01 Jan 2026	1.6 MB
03 Jan 2026	1.6 MB

‹ ▶ ›

It finds a mover, and names it.

Signal-to-noise at the field centre, visit by visit



Fifteen D2 visits of one spot. Fourteen are background, SNR 0.5 to 2.4. On 25 Dec 2025 it is 374, and SkyBoT places asteroid (152) Atala, V 13.8, 2.3″ from the centre at that exposure’s own time.

15 D2 visits over 369 days; SkyBoT, IMCCE.

It refuses a false discovery.

On 6 Jul 2025 the same spot flashed at SNR 140, and no catalogue explained it. It was one detector pixel the SPHEREx pipeline had already flagged as a cosmic-ray hit. Muninn reads those flags and masks it before measuring.

Atala’s own pixels on 25 Dec carry only OUTLIER, and spread like a real source.

6 Jul 2025, the centre of the Atala field

SNR without the flags

~~140~~

one pixel at 11× its neighbours

SNR with the flags read

1.2

background, like the other visits

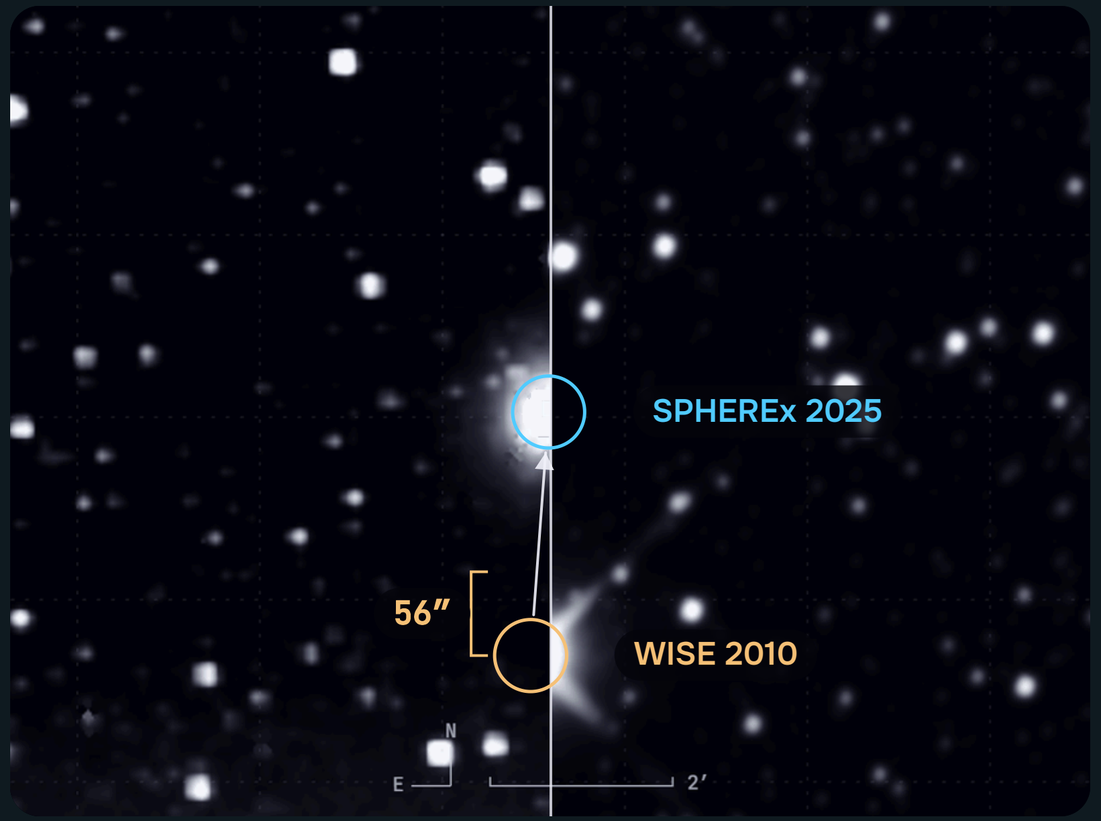
FLAGS	bit 0	bit 2	bit 6
0x45	TRANSIENT	SUR_ERROR	NONFUNC

The pipeline’s own label for that pixel: TRANSIENT — its signal jumped while the detector was read, as a cosmic ray does; SUR_ERROR — the readout could not be fitted; NONFUNC — not usable.

Fifteen years on, Barnard’s Star has moved about 157″ from where WISE saw it.

B of the wipe can be the AllWISE picture at 3.4 or 4.6 µm, on the same grid as a SPHEREx visit. The idea is WiseView’s, the blink tool the challenge names.

AllWISE Atlas through NASA IRSA. Markers: the star’s peak on each picture.



A SPHEREx, D2, 21 Aug 2025

B AllWISE W1, 3.4 µm, 2010–2011

≈157″ in 15.1 years

Barnard’s Star since WISE, 10.4″ × 15.1 years

56″

from its 2016 place to the WISE peak

2.5″

from Gaia carried back to 2010

10.4″ a year

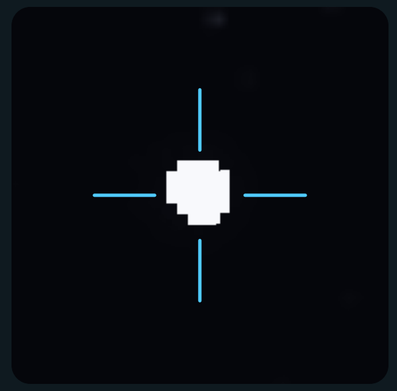
Barnard’s Star’s proper motion

The challenge’s five test targets, one press each.

Each opens in the band that suits it. The crosshair marks the target on a real SPHEREx frame.

Visit counts from NASA IRSA, SPHEREx DR1 and QR3, 13 Nov 2026.

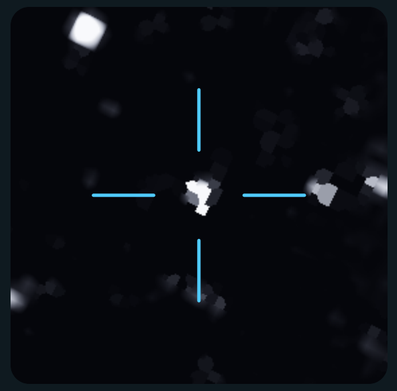
Five targets named by the challenge



Barnard's Star

Fastest star in the sky, 10.4″ a year

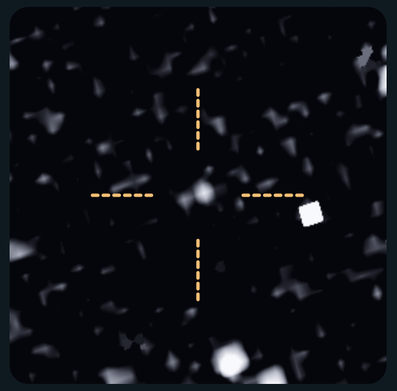
`D2 · 17 visits · 397 d`



ε Indi B

A pair of brown dwarfs beside ε Indi A

`D2 · 29 visits · 526 d`



WISE 0855−0714

Coldest known brown dwarf

`D6 · 17 visits · 370 d`

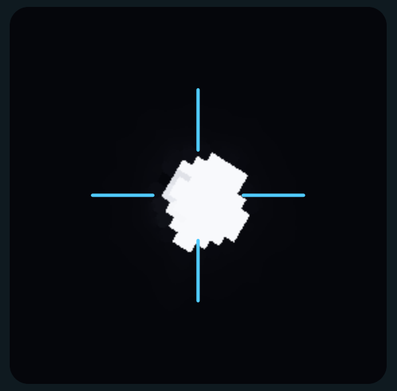
Too faint for one frame: about 17.2 AB against a 16.7 AB limit in D6. The crosshair is the challenge’s position.



RW Aurigae

A young variable star

`D2 · 16 visits · 372 d`



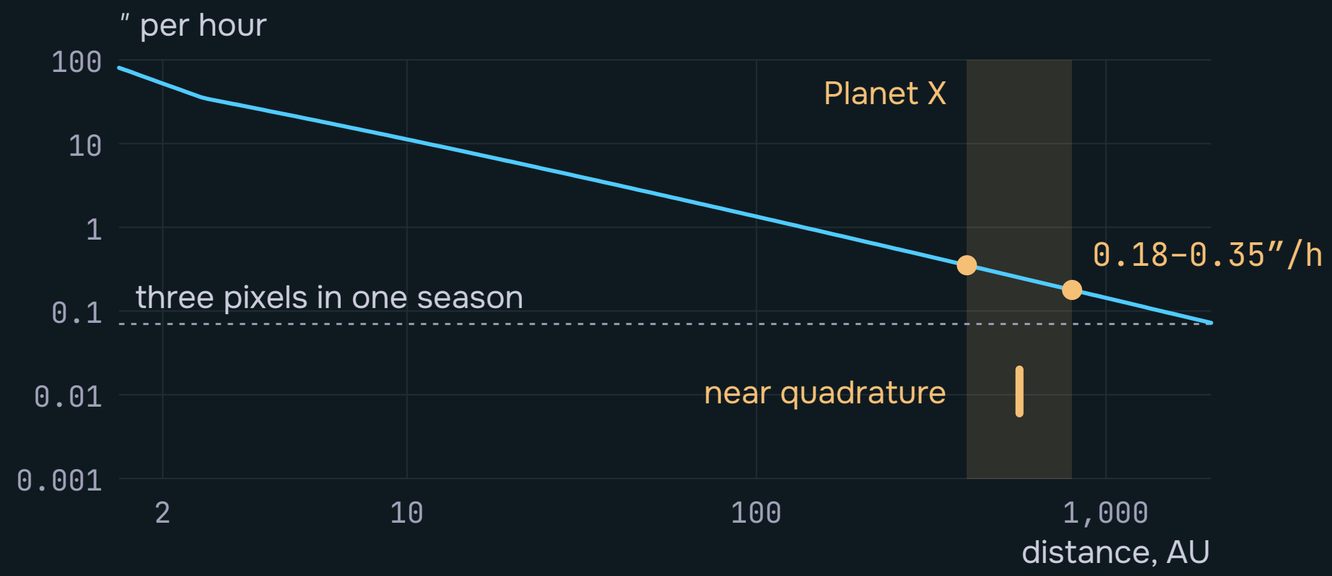
R Draconis

A Mira variable near the north ecliptic pole

`D3 · 55 visits · 414 d`

At 400–800 AU the motion test within a season is blind; the season link checks the parallax arc.

Apparent rate against distance from the Sun



Inside one season Hunter X needs three pixels in eleven days, 0.07″ an hour. SPHEREx sees Planet X near quadrature, at about 0.006–0.02″ an hour: it would read as fixed.

Curve computed on this slide: the fastest a body at that distance moves, at opposition.

The season link, one season to the next

1. A body is put on the mark’s line of sight at each distance from 30 to 2000 AU.
2. The Earth’s move across the Sun throws it along an arc in the next season: 13.6′ at 500 AU.
3. A source on the arc counts only if that spot was empty in every visit of the first season.

On a synthetic two-season sky, a source placed at 300 AU links at 300 AU, range 269–321.

Even 200 visits added would leave SPHEREx 2.1 to 3.8 magnitudes short of Planet X.

Measured from delivered detector pixels: 17.61 AB for one visit, 5.0 to 6.7 magnitudes short. Two hundred visits, extrapolated as √N, reach 20.49: still short, and 1.2 short even of the bright 21.7 case.

Prediction: Linder & Mordasini 2016, A&A 589, A134, V 22.6–24.3 near aphelion.

Faintest point source at 5σ, AB magnitudes



one visit, measured

17.61 AB

200 visits, extrapolated as √N

20.49 AB

Assumed: the V-band prediction against SPHEREx AB depth at 1.1–1.6 μm, a fair best case for a sunlike reflector.

Built on the challenge's resources, each one named.

Every service is asked live; the page stores no pixels. Muninn is not affiliated with NASA.

Challenge resources · NASA

NASA IRSA · SPHEREx
Every pixel, DR1 and QR3; the frame index; the FLAGS plane

SPHEREx Tutorial Notebooks
The reference for the DR1 file layout

WiseView
The model for the WISE wipe

Gaia DR3 `MAST, STScI`
What a marked star is, in a second or two

Challenge resources · CSA

CADC
Which other telescopes looked at a field

NEOSSat
Muninn shows NEOSSat's newest raw frame of a field, from CADC

Also used

ESA Gaia / DPAC
Star positions over the frame, moved to the visit date

NASA WISE / NEOWISE
AllWISE pictures; CatWISE2020 motions

2MASS
J, H and K of a marked source

IMCCE SkyBoT
Known asteroids at each exposure

JPL SSD
Small-Body Identification, beside SkyBoT

CDS
SIMBAD, Sesame and VizieR

Open it and blink a field yourself.

Live `muninn.mydevbox.xyz`

Code `github.com/root3315/muninn`

Team Yggdrasil · Uzbekistan

Khurshiddinov Timur Khurshiddinovich (team owner)

Stepan Ionichev

Edem Galiev

Khamidkhodjayeva Fayziya

Muhammadyusuf Bahtiyarov

Temur Kholjigitov

Does today

- Blink, wipe, difference and season stack on one north-up grid
- Measures a marked spot in every visit, and links it across seasons
- Names known asteroids; lays Gaia and WISE over the frame
- States each field’s 5σ limit
- Compares brightness at one wavelength across visits

Does not

- Reach Planet X: 2.1–3.8 mag short even with 200 visits
- Show a Gaia image: it is a catalogue, so the wipe draws a model from it
- Find faint movers on its own: Hunter X ranks candidates, and an injection test did not recover faint ones

Next: the full 102-channel spectrum, visits added along a trial orbit for fainter bodies, and every season linked, not only the nearest.

Appendix

How each part works, for questions.

1 Enter through the real sky.

2 Five ways to compare one patch of sky.

3 Six bands, six field sizes, one link.

4 A 65 MB file, read a megabyte or two at a time.

5 Click a spot: it is measured in every visit.

6 Gaia and WISE, laid over the frame.

7 Which other telescopes looked at the same field.

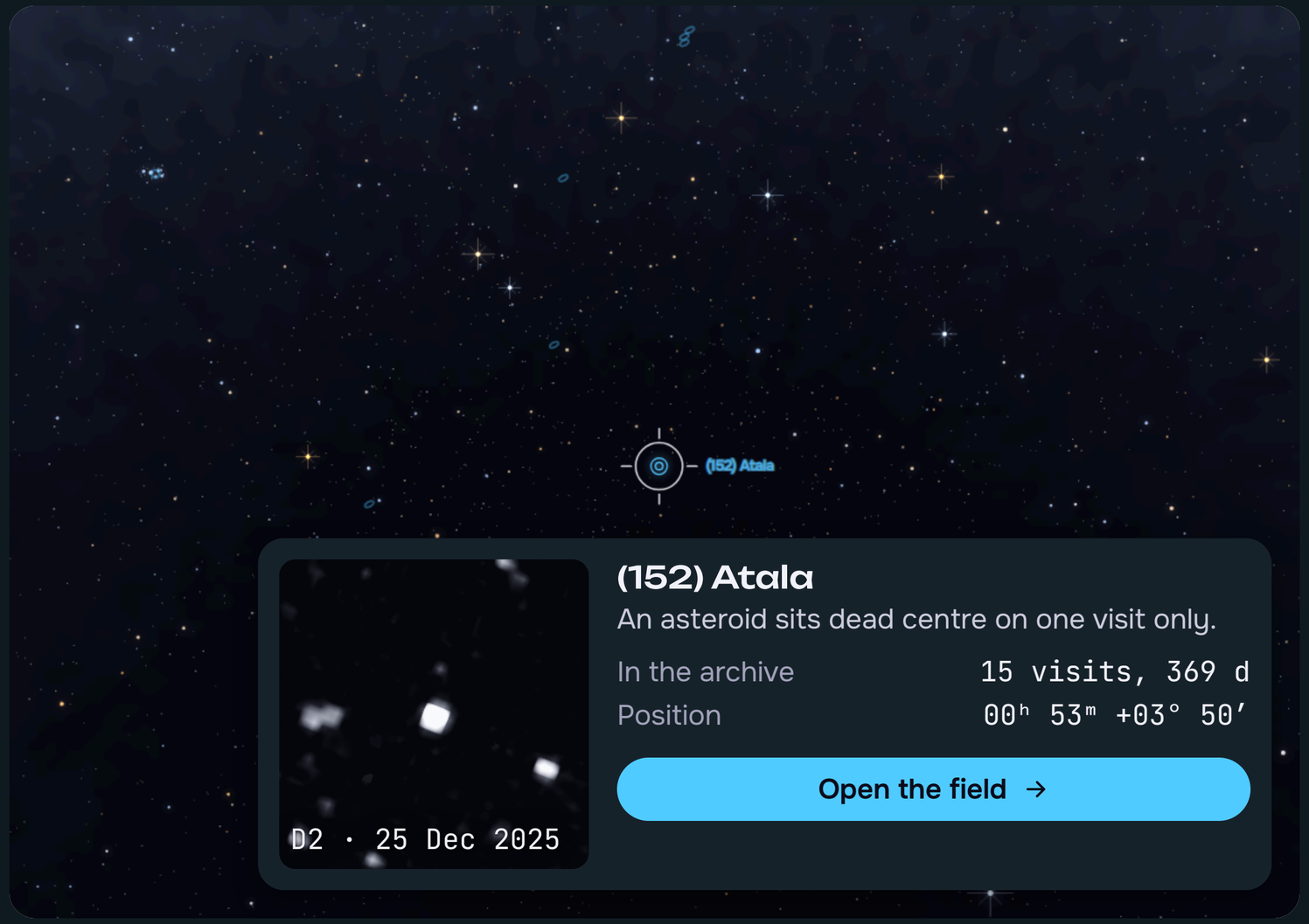
8 A new SPHEREx release is one line.

Enter through the real sky.

The first screen is the sky: 15,568 catalogued stars, turning. Press a star and a reticle closes on it; a card brings a real SPHEREx frame of that place.

1. The sky turns
2. A press: the reticle closes
3. The card: a real SPHEREx frame
4. The dive into the field

Drawn by the tool’s own sky code. Bright Star Catalogue and Hipparcos.

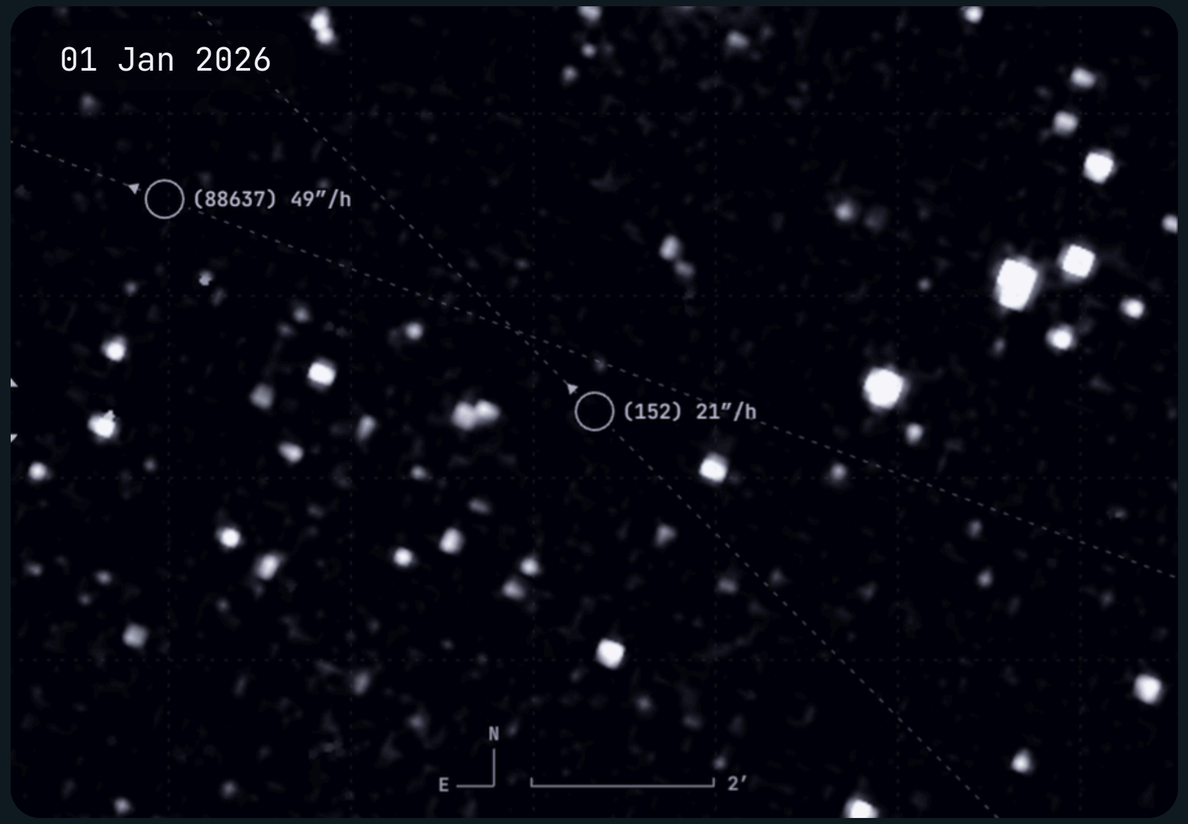


Five ways to compare one patch of sky.

Every visit is resampled onto one north-up grid. A is 25 Dec 2025 with (152) Atala ringed; B is 1 Jan 2026, when it has gone.

The Atala field, D2, as the tool draws it.

Blink | Wipe | Difference | Single | Stack



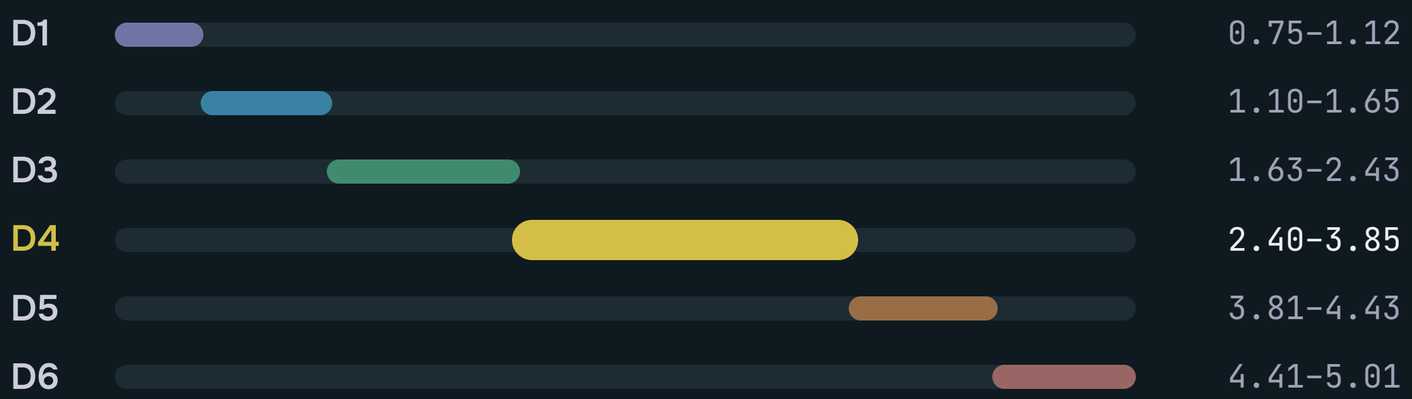
Every visit in sequence — the quickest way to catch something that shifted

Six bands, six field sizes, one link.

Six bands, D1 to D6, and the page takes the hue of the band in the beam. The field runs 6′ to 36′; the address carries place, size, band and language.

Band edges read from the Atala field’s frame index.

Wavelength, µm



Field size, short side

6′ 9′ 12′ 18′ 27′ 36′

The address

```
https://muninn.mydevbox.xyz/#ra=13.41940&dec=3.84590&fov=0.3&band=D4
&lang=en
```

A 65 MB file, read a megabyte or two at a time.

The browser asks the public AWS mirror for byte ranges: the header, the detector rows the field covers, and their flags. No pixels pass through our server.

65.4 MB per Level-2 frame; bytes measured on slide 2, as it reads.

2.4% of each file is read

What the four frames on slide 2 cost

24 Dec 2025	1.6 MB
25 Dec 2025	1.6 MB
01 Jan 2026	1.6 MB
03 Jan 2026	1.6 MB

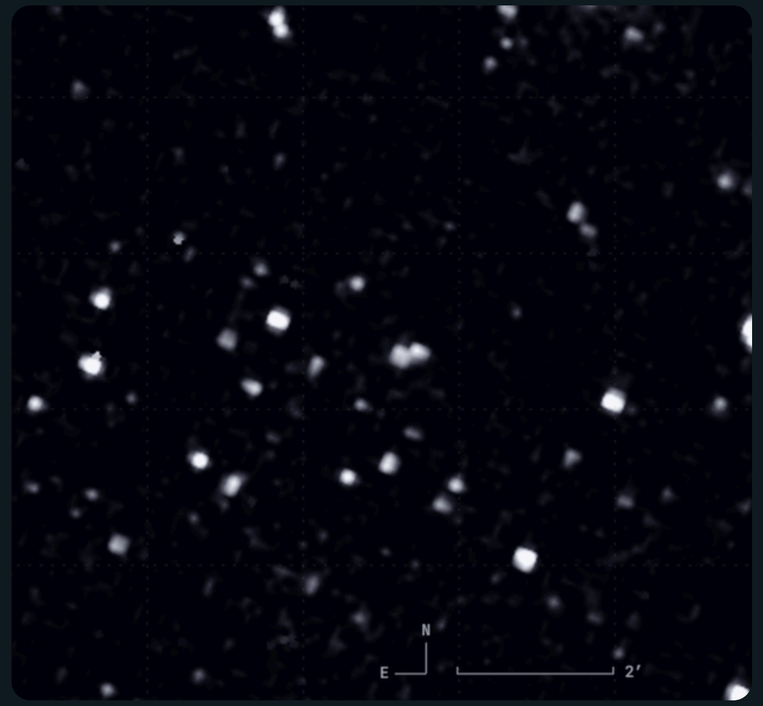
Read 6.4 MB of 262 MB held at the archive: 2.4%

64 KB	FITS header and the WCS: where each pixel is on the sky
1–3 MB	only the detector rows the field covers
1–3 MB	the same rows of the FLAGS plane

Click a spot: it is measured in every visit.

Position, AB magnitude in a 3×3-pixel aperture, a verdict and what catalogues know, saved as CSV. One more press reads a spectrum: one point per exposure across D1–D6.

A star in the Atala field, marked in the tool.



1 **Fixed** ×

in all 10 visits, steady to 1.6″ — a fixed source

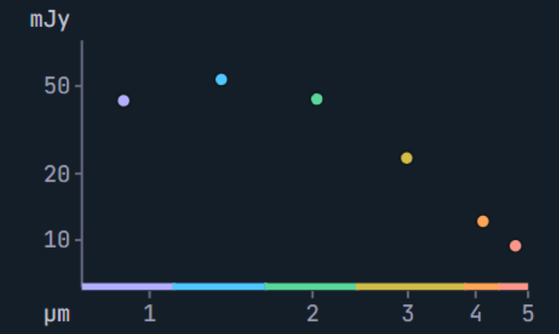
Gaia DR3: 2551684673789756800 — star, G 12.8ᵐ

2MASS · AllWISE: J 11.2 · H 10.6 · K 10.5 · W1 10.4 · W2 10.5

Gaia parallax 6.21 mas — about 161 pc away

CatWISE2020 (NASA WISE): moves 0.05″ a year

No answer from: SIMBAD, VSX



band	μm	mJy	date
D1	0.896	42.89 ± 0.07	6 Jul
D2	1.357	53.53 ± 0.07	6 Jul
D3	2.037	43.59 ± 0.06	7 Jul
D4	2.984	23.56 ± 0.04	6 Jul
D5	4.127	12.2 ± 0.1	6 Jul
D6	4.739	9.42 ± 0.19	7 Jul

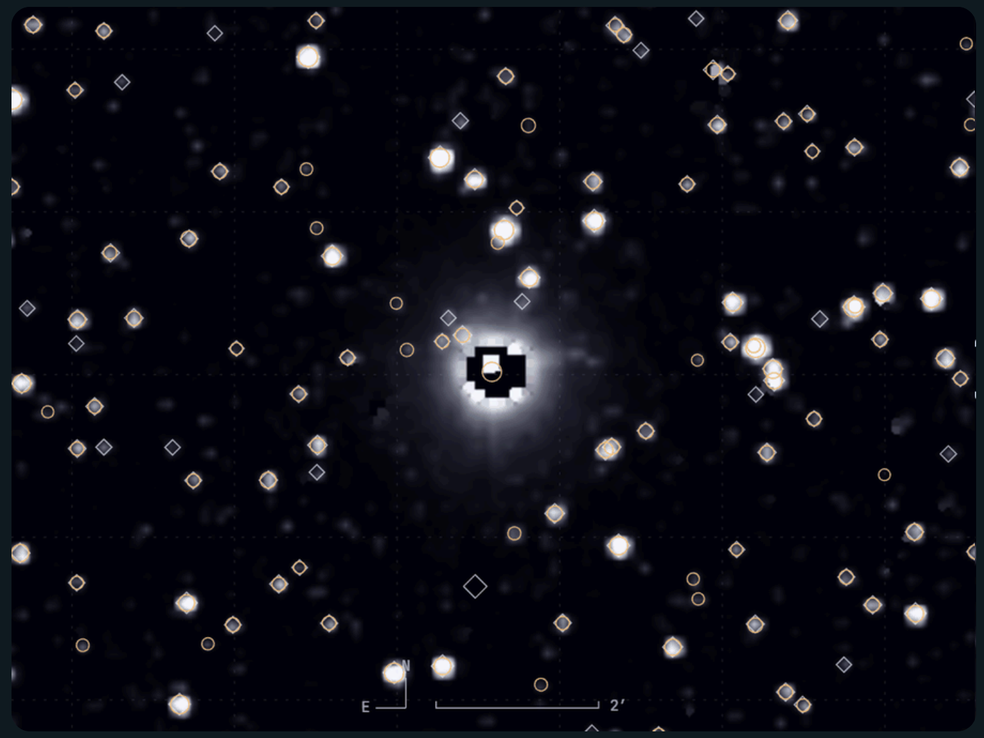
CSV columns

```
date_utc,mjd,obs_id,band,wavelength_um,bandwidth_um,ra_deg,dec_deg,result,flux_mjy,flux_err_mjy,limit_5sigma_mjy,mag_ab,peak_snr
```

Gaia and WISE, laid over the frame.

Gaia DR3 positions are moved from 2016 to the date of the visit; AllWISE positions stay where WISE saw them in 2010.

Barnard’s Star, D2, 12′ × 9′; its core masked by the pipeline’s flags.



- ○ Gaia DR3 (ESA): moved to the date of this visit
- ◇ AllWISE (NASA WISE): where they were in 2010

Asked for a marked source

Gaia DR3 `NASA MAST`
what the star is, in a second or two

Gaia DR3 · AllWISE · 2MASS `NASA IRSA · CDS VizieR`
the same catalogues raced: the first answer is used

CatWISE2020 `NASA IRSA`
a proper motion from ten years of WISE, said only at five times its error

SIMBAD `CDS`
what it is called

VSX `AAVSO`
whether it is a known variable

Which other telescopes looked at the same field.

For every field the page asks the Canadian Astronomy Data Centre how many science observations each archive holds, and shows NEOSSat's newest raw frame of it.

CADC, within 0.1° of Barnard's Star; log scale.

In other archives: Barnard's Star



A new SPHEREx release is one line.

IRSA publishes each release as its own collection, and the page names them in one list. On 13 Nov 2026 it moved from Quick Release 2 to DR1 and QR3.

The page stores no pixels, so new data need no rebuild.

public/js/release.js

```
~~export const COLLECTIONS = ['spherex_qr2', 'spherex_qr2_deep'];~~
export const COLLECTIONS = ['spherex_dr1', 'spherex_dr1_deep',
'spherex_qr3', 'spherex_qr3_deep'];
```

13 Nov 2026	the list of collections changes
npm run bake	the ten ready fields get new frame indexes
npm test	checked offline, then deployed